CS 412 / 512
Introduction to Computer Graphics

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COMPUTER SCIENCE

3D Transformation

3D Representation

2D point $(x, y, 1)$ → 3D point $(x, y, z, 1)$

2D vector $[x\ y\ 0]^T$ → 3D vector $[x\ y\ z\ 0]^T$

Homogeneous Coordinates

2D

$$\begin{bmatrix} a & b & c \\ d & e & f \\ 0 & 0 & 1 \end{bmatrix}$$

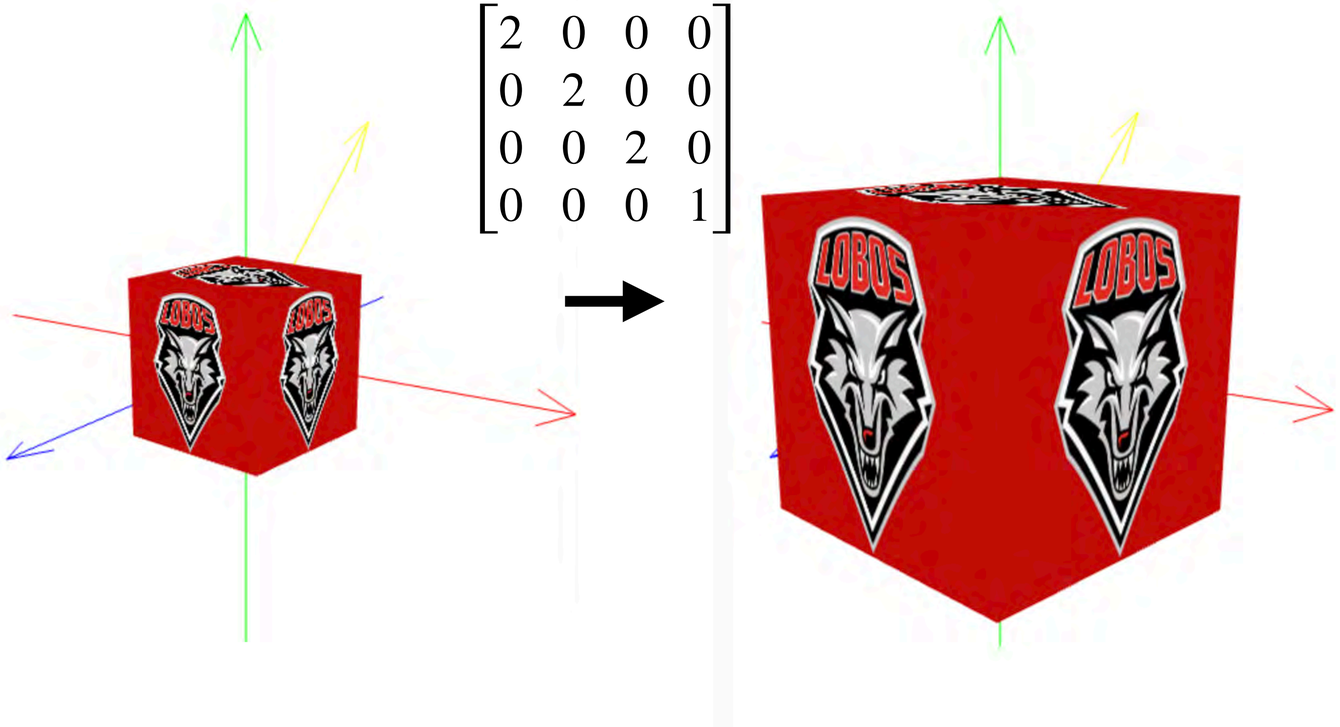
→

3D

$$\begin{bmatrix} a & b & c & d \\ e & f & g & h \\ i & j & k & l \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

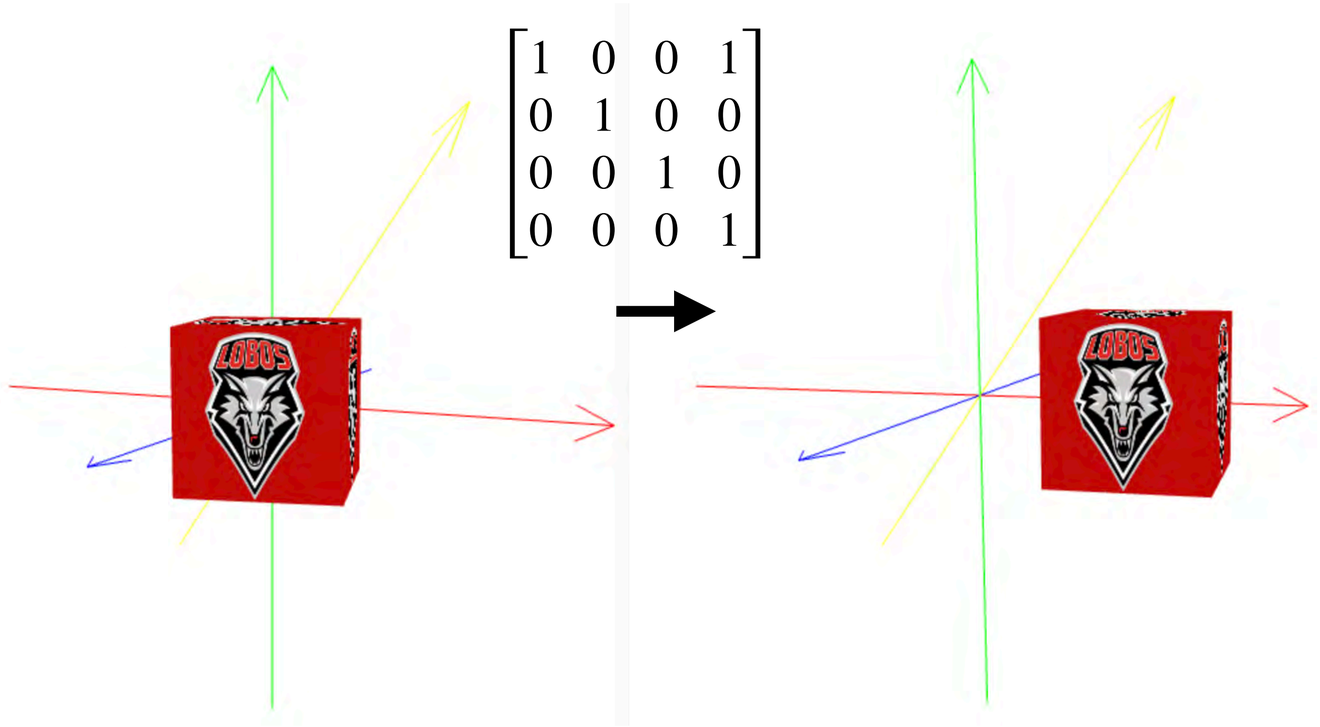
3D Scaling

$$\begin{bmatrix} S_x & 0 & 0 \\ 0 & S_y & 0 \\ 0 & 0 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} S_x & 0 & 0 & 0 \\ 0 & S_y & 0 & 0 \\ 0 & 0 & S_z & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$



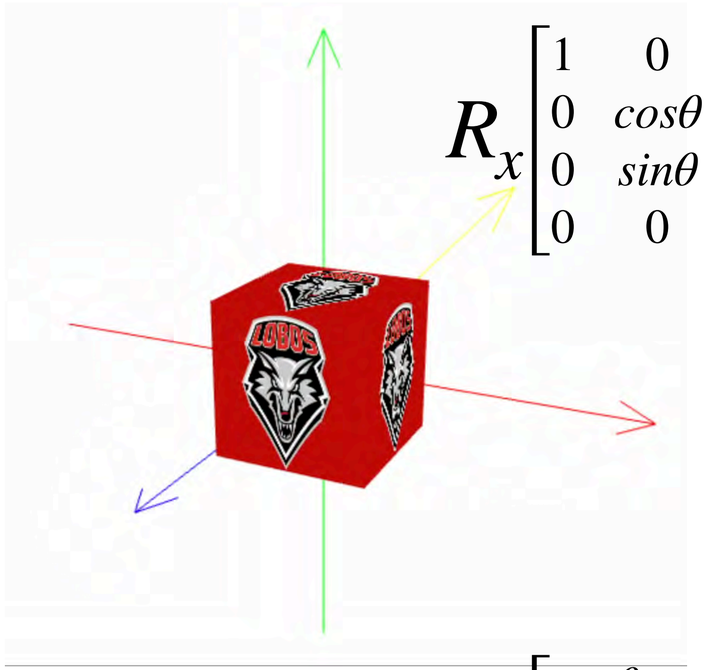
3D Translation

$$\begin{bmatrix} 1 & 0 & t_x \\ 0 & 1 & t_y \\ 0 & 0 & 1 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0 & 0 & t_x \\ 0 & 1 & 0 & t_y \\ 0 & 0 & 1 & t_z \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

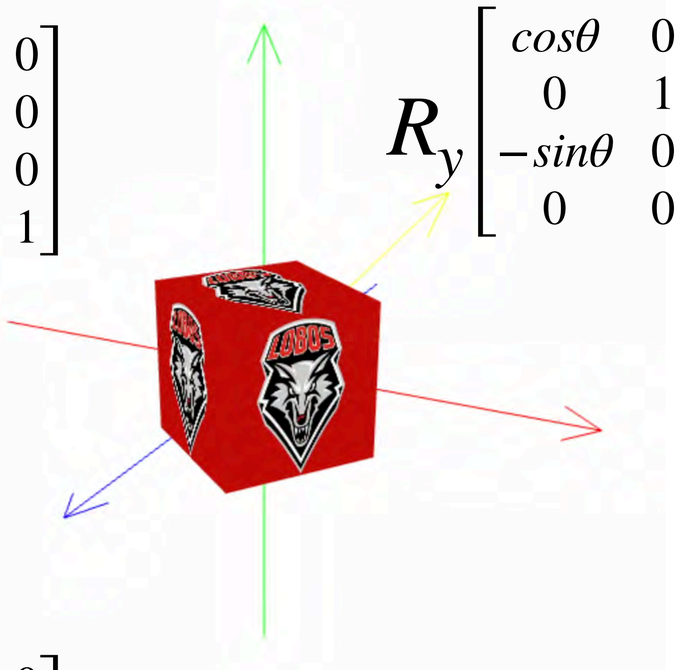


3D Rotation

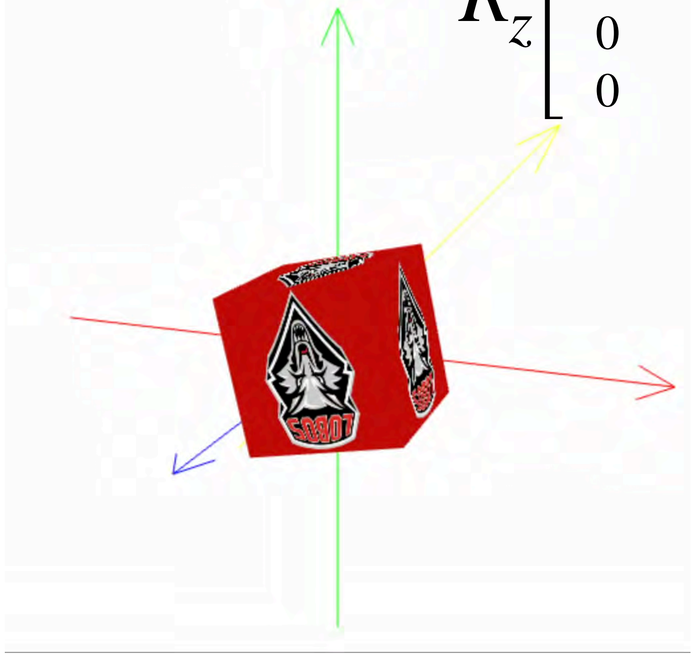
$$\begin{bmatrix} cos\theta & -sin\theta & 0 \\ sin\theta & cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$



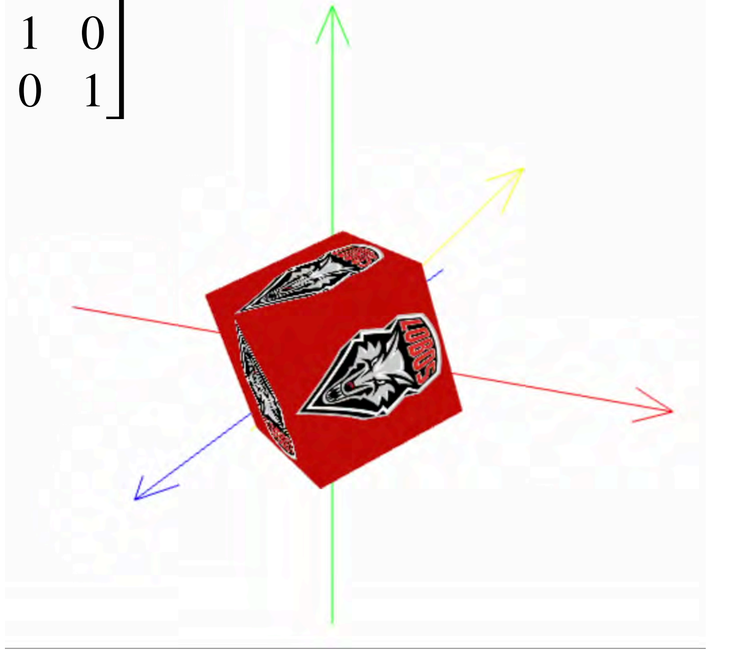
$$R_x \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & cos\theta & -sin\theta & 0 \\ 0 & sin\theta & cos\theta & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$



$$R_y \begin{bmatrix} cos\theta & 0 & sin\theta & 0 \\ 0 & 1 & 0 & 0 \\ -sin\theta & 0 & cos\theta & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$



$$R_z \begin{bmatrix} cos\theta & -sin\theta & 0 & 0 \\ sin\theta & cos\theta & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$



General Rotation

Rodrigues' rotation formula

If rotate θ around an arbitrary unit vector $\mathbf{u} = (u_x, u_y, u_z)$

$$R = \begin{bmatrix} \cos\theta + (1-\cos\theta)u_x^2 & (1-\cos\theta)u_x u_y - u_z \sin\theta & (1-\cos\theta)u_x u_z + u_y \sin\theta \\ (1-\cos\theta)u_x u_y + u_z \sin\theta & \cos\theta + (1-\cos\theta)u_y^2 & (1-\cos\theta)u_y u_z - u_x \sin\theta \\ (1-\cos\theta)u_x u_z - u_y \sin\theta & (1-\cos\theta)u_y u_z + u_x \sin\theta & \cos\theta + (1-\cos\theta)u_z^2 \end{bmatrix}$$

General Rotation

Representation as a sequence of rotations around coordinate axes

$$R = R_z(\gamma)R_y(\beta)R_x(\alpha) \qquad R_z(\gamma)R_y(\beta)R_x(\alpha) \neq R_x(\alpha)R_y(\beta)R_z(\gamma)$$ or other permutations

$$R = \begin{bmatrix} cos\gamma & -sin\gamma & 0 \\ sin\gamma & cos\gamma & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} cos\beta & 0 & sin\beta \\ 0 & 1 & 0 \\ -sin\beta & 0 & cos\beta \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & cos\alpha & -sin\alpha \\ 0 & sin\alpha & cos\alpha \end{bmatrix}$$

$$= \begin{bmatrix} cos\gamma\ cos\beta & cos\gamma\ sin\beta\ sin\alpha - sin\gamma\ cos\alpha & cos\gamma\ sin\beta\ cos\alpha + sin\gamma\ sin\alpha \\ sin\gamma\ cos\beta & sin\gamma\ sin\beta\ sin\alpha + cos\gamma\ cos\alpha & sin\gamma\ sin\beta\ cos\alpha - cos\gamma\ sin\alpha \\ -sin\beta & cos\beta\ sin\alpha & cos\beta\ cos\alpha \end{bmatrix}$$

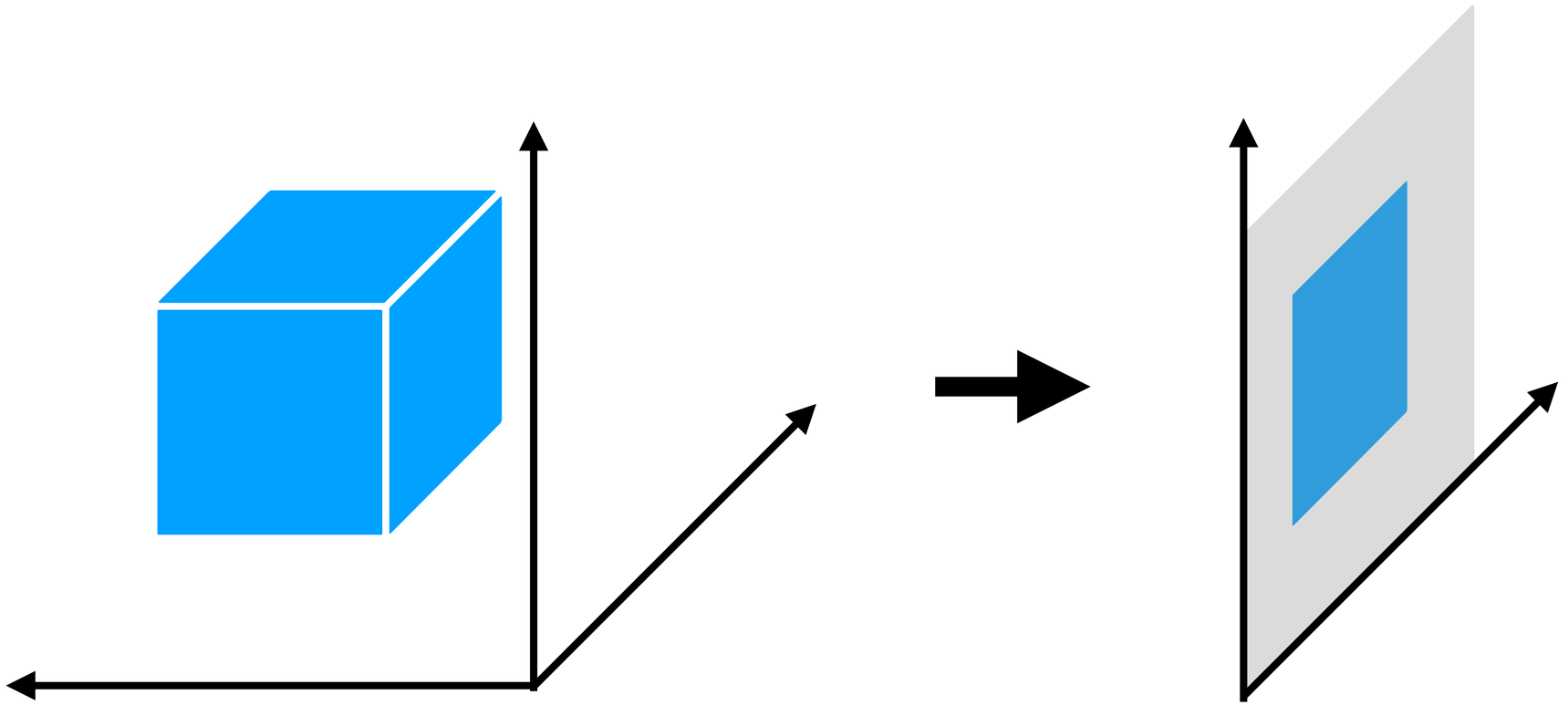
Scaling, Rotation, Translation

The most common pipeline

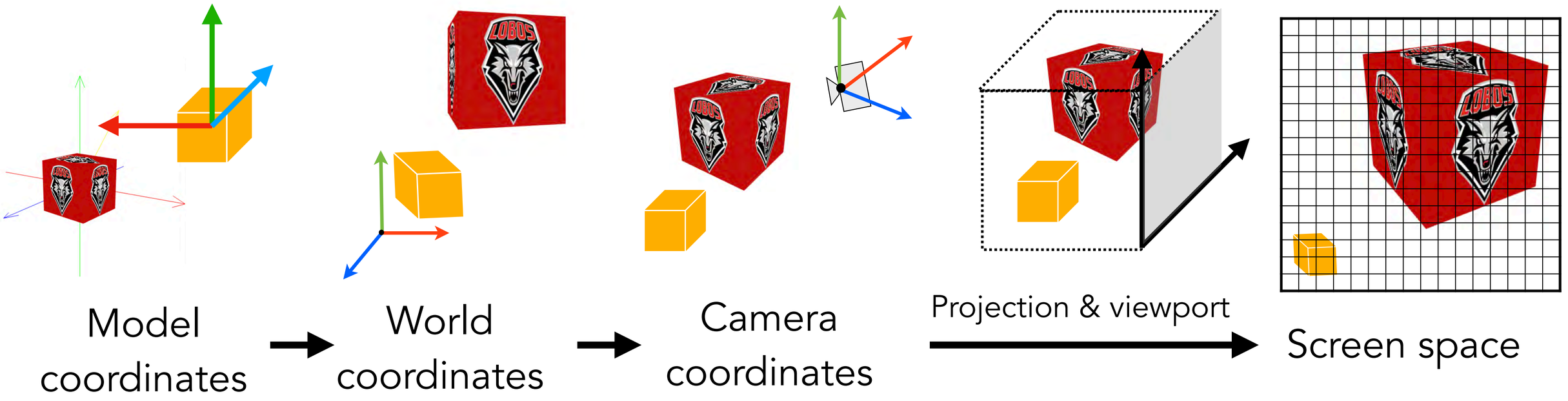
The final transformation matrix $M = T \cdot R \cdot S$

$$p' = M \cdot p = T \cdot R \cdot S \cdot p$$

Viewing Transformations

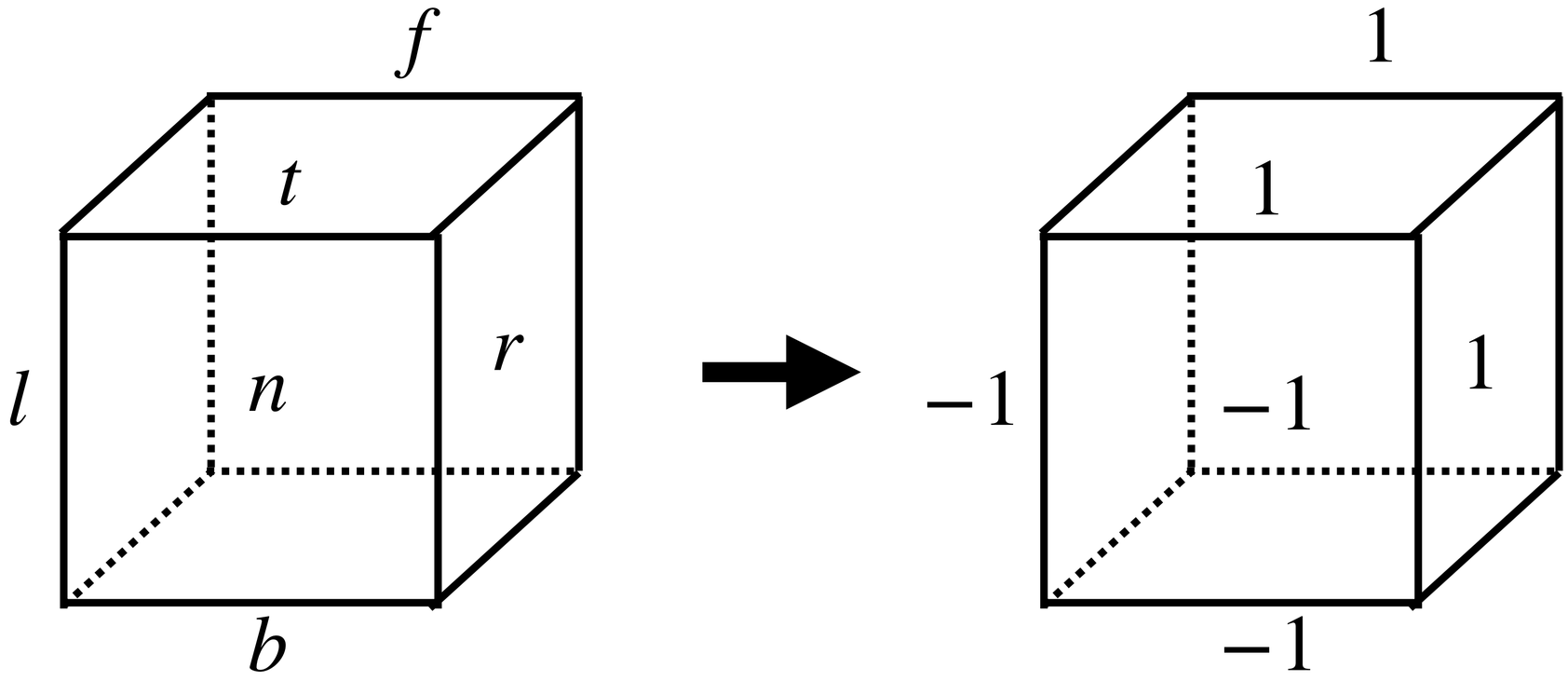


Viewing Transformations



Projection Transformations

Orthographic Projection



Camera Space

Canonical View Space

Projection Transformations

Orthographic Projection Matrix

$$\begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} \frac{2}{r-l} & 0 & 0 & -\frac{r+l}{r-l} \\ 0 & \frac{2}{t-b} & 0 & -\frac{t+b}{t-b} \\ 0 & 0 & \frac{2}{f-n} & -\frac{f+n}{f-n} \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

Projection Transformations

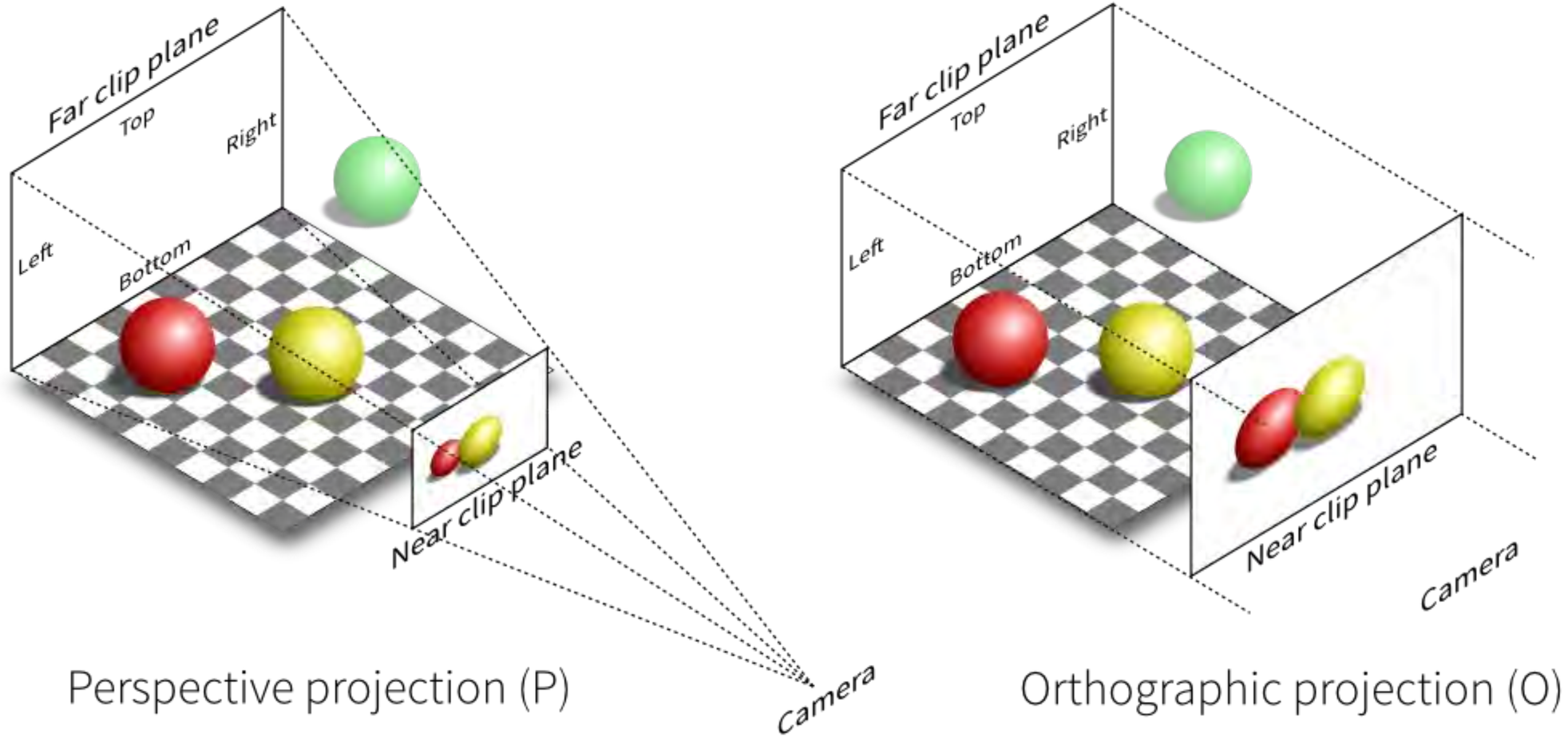
Orthographic vs Perspective Projection





Projection Transformations

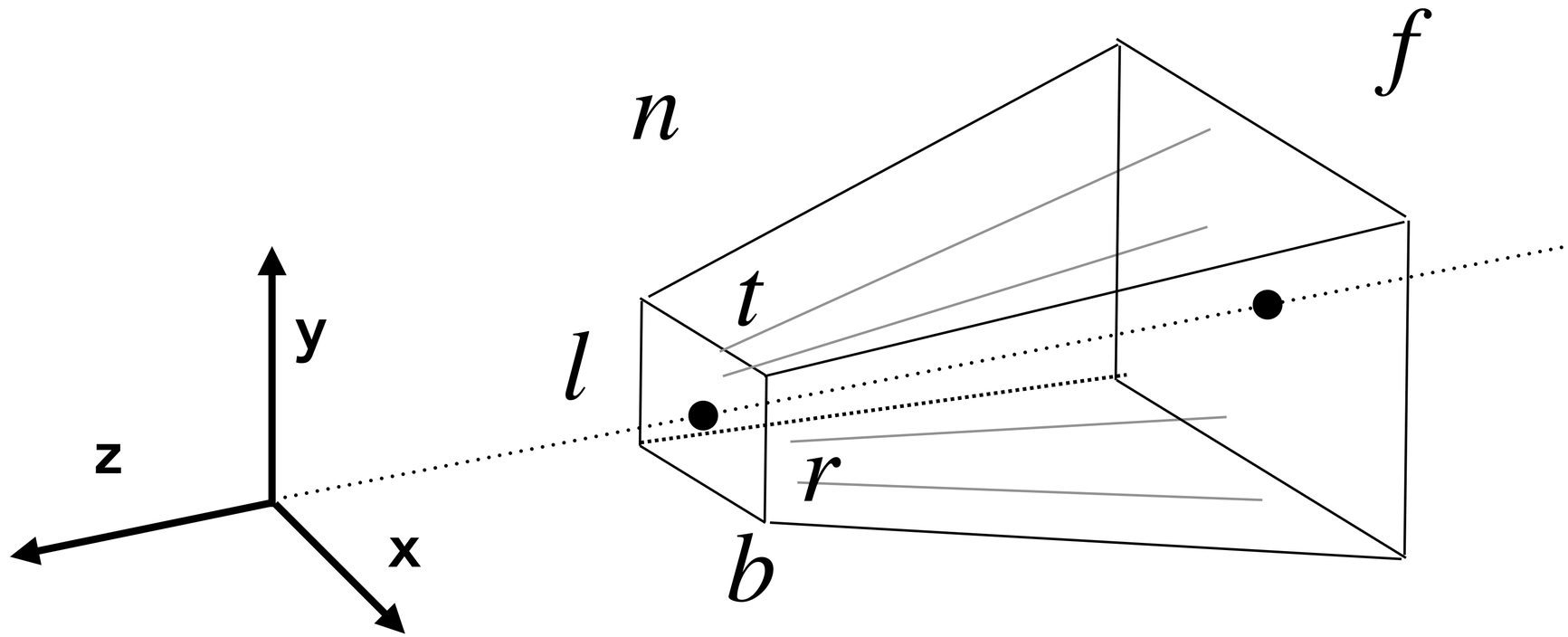
Orthographic vs Perspective Projection



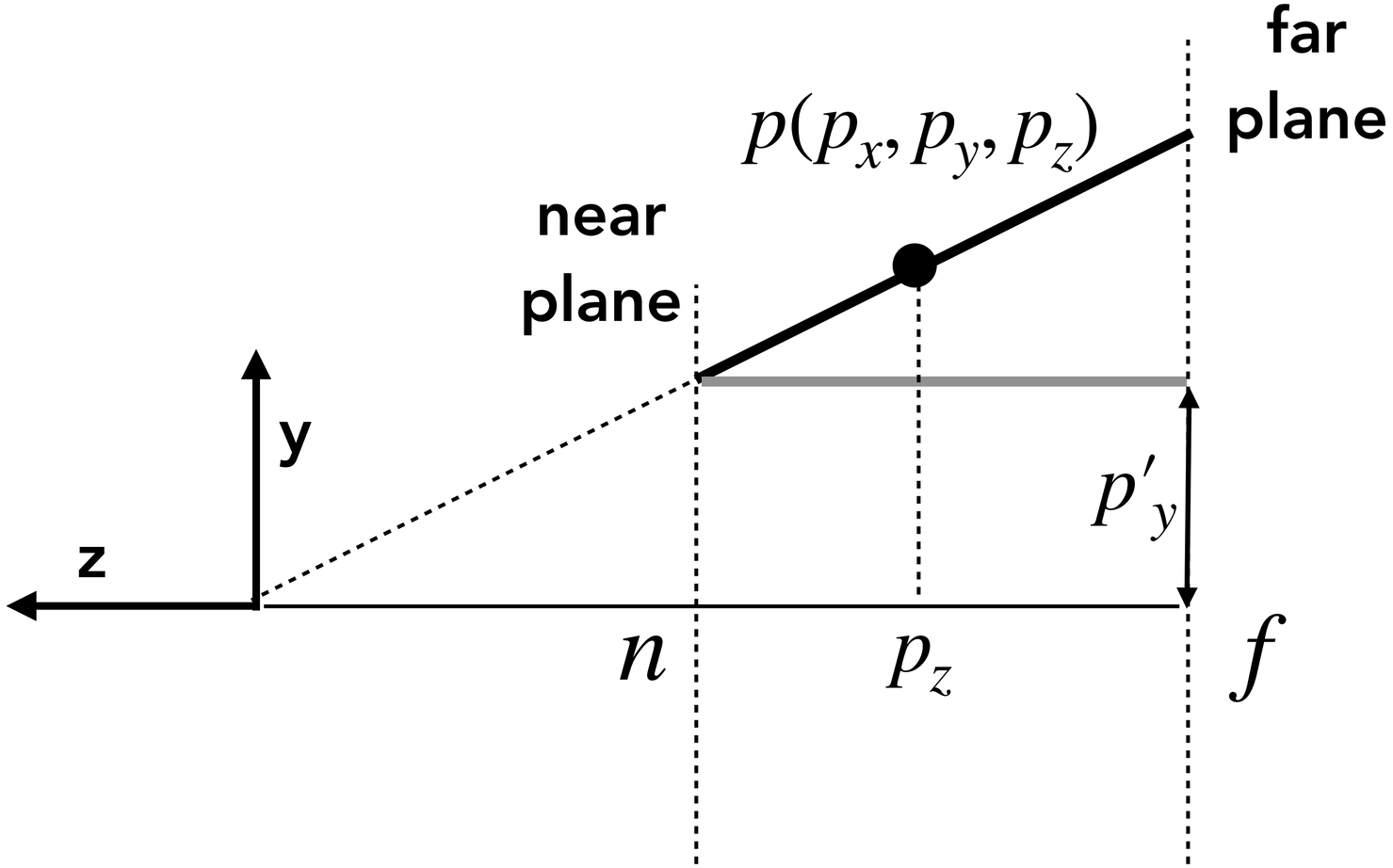
[source: Oleg Varaksin]

Perspective Projection

Symmetric Frustum



Perspective Projection



$$\frac{p_y}{p_z} = \frac{p'_y}{-n}$$

$$p'_y = \frac{-n}{p_z} p_y$$

Same for z-x plane

$$p'_x = \frac{-n}{p_z} p_x$$

Perspective Projection Matrix

$$[x\ y\ z\ 1]^T \longrightarrow [x'\ y'\ z'\ w]^T$$

$$x' = nx$$
$$y' = ny$$
$$w = -z$$
$$z' = Az + B$$

$$\begin{bmatrix} x' \\ y' \\ z' \\ w \end{bmatrix} = \begin{bmatrix} n & 0 & 0 & 0 \\ 0 & n & 0 & 0 \\ 0 & 0 & A & B \\ 0 & 0 & -1 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

Perspective Projection Matrix

Mapping z to standard normalized [-1, 1]

$$z' = \frac{Az + B}{-z} = \begin{cases} -1 & if\ z = -n \\ 1 & if\ z = -f \end{cases}$$

$$\begin{bmatrix} x' \\ y' \\ z' \\ w \end{bmatrix} = \begin{bmatrix} n & 0 & 0 & 0 \\ 0 & n & 0 & 0 \\ 0 & 0 & -\frac{f+n}{f-n} & -\frac{2fn}{f-n} \\ 0 & 0 & -1 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

Simplified mapping to [n, f]

$$z' = \frac{Az + B}{z} = \begin{cases} n & if\ z = n \\ f & if\ z = f \end{cases}$$

$$\begin{bmatrix} x' \\ y' \\ z' \\ w \end{bmatrix} = \begin{bmatrix} n & 0 & 0 & 0 \\ 0 & n & 0 & 0 \\ 0 & 0 & f+n & -fn \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

Perspective Projection

Asymmetric Frustum Matrix

$$M = \begin{bmatrix} \frac{2n}{r-l} & 0 & \frac{r+l}{r-l} & 0 \\ 0 & \frac{2n}{t-b} & \frac{t+b}{t-b} & 0 \\ 0 & 0 & -\frac{f+n}{f-n} & -\frac{2fn}{f-n} \\ 0 & 0 & -1 & 0 \end{bmatrix}$$

Quiz 1